AUTONOMOUS DATA FERRYING FROM A SELF-ORGANIZING/HEALING WSN IN DISCONNECTED ZONES

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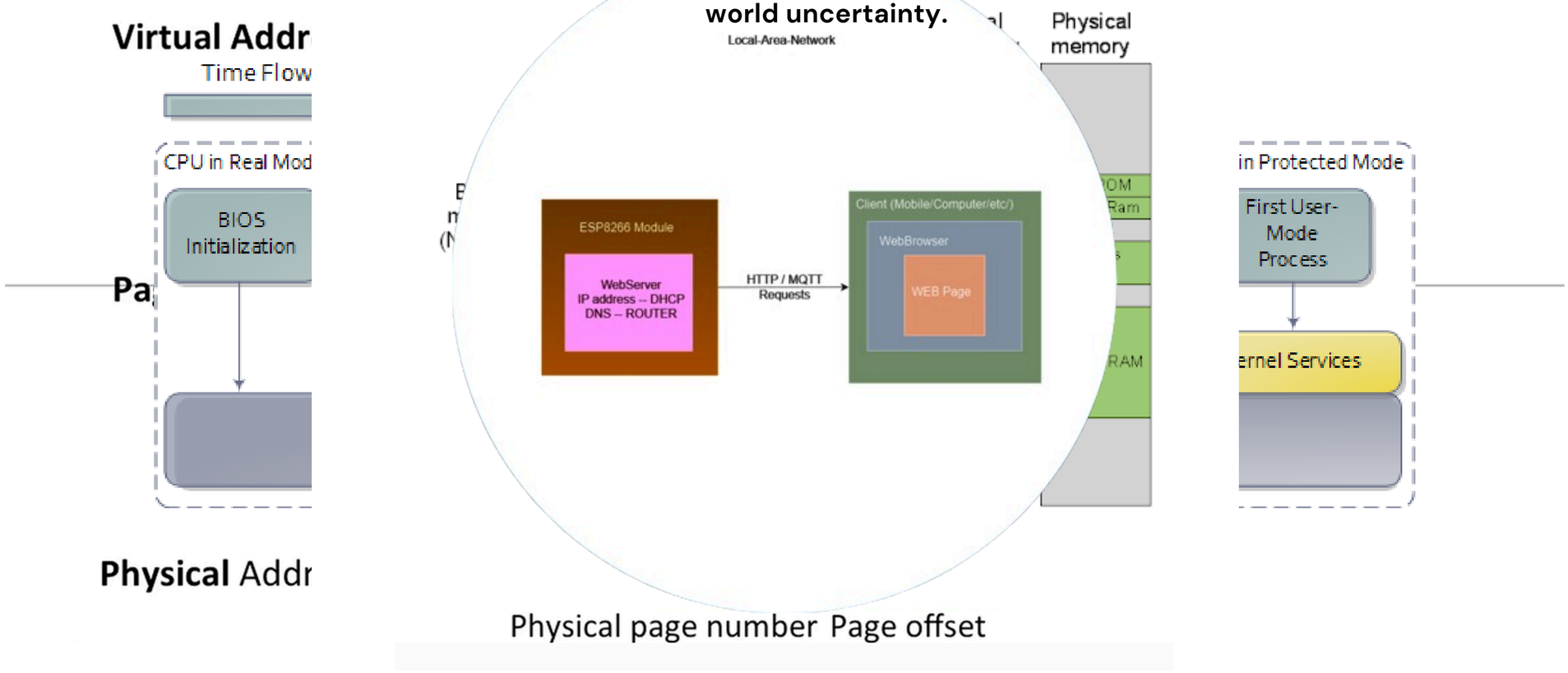
Individual Component Overview

Wireless Sensor Networks (WSNs) operating in infrastructure less, dynamic, and resource constrained environments require autonomous mechanisms to select reliable and energy efficient Cluster Heads (CHs). Traditional clustering and leader election approaches primarily rely on static weighting, single metric optimisation, or predefined thresholds, which are insufficient under real world uncertainty.



Individual Component Overview

How STELLAR Solves the Problem

- Raw sensing data is stabilised using EWMA
- Metric instability is explicitly quantified
- Noisy metrics automatically lose influence
- Weights adapt safely over time
- Leader selection balances fairness and performance

Key Requirements Addressed

Functional Requirements

- Autonomous CH election (fully decentralized)
- Dynamic metric adaptation
- Multi metric decision making
- Fair leader selection (Pareto + Nash)
- Robust re-election handling

Non-Functional Requirements

- **Energy Efficiency**
- Penalizes low battery nodes
- Reduces unnecessary re-elections

- **Stability & Convergence**
- Lyapunov based adaptation
- Smooth, non oscillatory behavior

- **Robustness & Security**
- Resistant to noisy measurements
- Trust integrates HMAC + packet reliability
- Malicious or unstable nodes lose leadership chance

Design Overview and Design Contribution

Key Design Contributions

Stability Aware Metric Evaluation

- Separates metric value from metric reliability
- Uses EWMA + variance + entropy to down-weight noisy metrics
- Topology Aware Centrality: Spatial variance of neighbor RSSI to prefer geographically central leaders for better coverage.

Adaptive Weight Learning

- Metric importance is learned online
- Lyapunov inspired update ensures smooth, stable convergence

Non-Linear Utility Modelling

- Realistic utility curves for battery, trust, uptime, and link quality
- Prevents misleading linear aggregation

System Design and Prototype Status

System Design Structure

- **Hardware Inputs: Battery ADC, system timer, RSSI, packet sequence numbers**
- **Processing Layers:**

- Metric normalization (0-1)
- EWMA smoothing
- Variance & entropy calculation
- Adaptive weight update
- Final STELLAR score computation

Design Overview and Design Contribution

Stability Awareness

$$\sigma_t^2 = \alpha(x_t - x_{t-1})^2 + (1-\alpha)\sigma_{t-1}^2$$

Uncertainty Modeling (Entropy)

$$H(m) = \frac{1}{2}\ln(2\pi e \cdot \sigma^2) \qquad c_i = \frac{e^{-\gamma H_i}}{\sum_j e^{-\gamma H_j}}$$

Adaptive Weight Learning (Lyapunov Stability)

$$w_i^* = w_{\text{base},i}(1 + 0.5(c_i - 0.25)) \qquad w_i(t+1) = w_i(t) - \eta \nabla V$$

Fair Leader Selection (Nash Bargaining)

$$n^* = \arg\max_{n \in P} \sum_i w_i \ln(\phi_i(n) - d_i)$$

Prototype Status

Fully implemented on ESP32 embedded nodes

Decentralized algorithm

Real-time execution within firmware loop

Implemented Components

Battery, uptime, trust, link quality pipeline

EWMA smoothing & uncertainty tracking

Adaptive weights & score computation

Cluster Head election logic

Next...

Formal Validation Through Controlled Experiments

Parameter Sensitivity & Time-Constant Analysis

Hardware Deployment Expansion to multiple clusters

ENERGY EFFICIENT AND FAULT TOLERANT CLUSTER DATA COMMUNICATION IN UAV ASSISTED WSN

COMPUTER SYSTEMS AND NETWORK ENGINEERING

Problem Focus and the proposed solution

- Addresses the efficient and reliable transfer of sensor data from Main Sets (MS) to the Cluster Head (CH) after CH election.
- Implements a scheduled, time-sliced MS → CH communication mechanism instead of continuous polling.
- Ensures temporary data storage at CH until UAV proximity is detected.
- Reduces energy consumption at MS nodes by minimizing idle listening and unnecessary transmissions.
- Supports scalability, allowing multiple MS nodes to operate without collisions.

Key Requirements Addressed by the Solution

- Reliable data collection from MS nodes without packet loss.
- Low-power operation suitable for battery-constrained embedded devices.
- Predictable communication behavior compatible with real-time embedded systems.
- Compatibility with ESP-based hardware and ESP-NOW communication.
- Minimal manual intervention once the system is deployed.

Design Excellence / Contribution

- Introduced slot-based scheduling to reduce idle listening.
- Clear functional separation between MS, and CH nodes.
- Embedded-oriented design considering memory, timing, and power limits.
- Modular architecture allowing future enhancements.

Feedback on Prototype

- Stable MS → CH communication achieved under scheduled transmission.
- Reduced active time observed at MS nodes compared to polling-based approaches.
- CH successfully buffers sensor data until UAV proximity is detected.
- Prototype validates the practical feasibility of the proposed design.

INTELLIGENT UAV-ASSISTED DATA FERRING IN DISCONNECTED SENSOR DEPLOYMENTS

COMPUTER SYSTEMS AND NETWORK ENGINEERING

Problem Focus

Problem Context

- Wireless Sensor Networks (WSNs) are often deployed in remote, large-scale, or disconnected environments
- Continuous connectivity using cellular or satellite links is:
  - Costly
  - Energy-inefficient
  - Sometimes infeasible

Core Challenges

- Intermittent or no network connectivity
- Limited sensor node power and storage
- Static data sinks lead to data loss and long latency
- Manual data collection is unsafe and inefficient

Research Gap

- Need for an autonomous, intelligent, and adaptive UAV-based data ferry
- Must operate reliably in disconnected and delay-tolerant networks
- Should integrate with real UAV flight stacks, not simulations alone

Proposed Solution

SKY-MULE: ROS-2 Based Intelligent UAV Data Ferrying System

Solution Overview

- ROS-FerryPilot is an autonomous UAV-based data ferrying system designed for disconnected environments
- The UAV acts as an intelligent mobile data mule, visiting sensor nodes, onboarding them securely, collecting data, and returning to base
- Built using:
  - ROS 2 for modular mission control
  - ArduPilot Copter 4.6.3 for reliable flight execution
  - MAVROS as the communication bridge between ROS 2 and the flight controller

Key Functional Components

Waypoint-Based Autonomous Navigation

- Mission waypoints are generated dynamically from a mission payload
- ArduPilot executes navigation in AUTO mode
- Supports takeoff, transit, hover, descent, and RTL behaviors

Mission-Specific ROS 2 Nodes

- Separate ROS 2 nodes handle different mission types, such as:
  - Waypoint mission management
  - WSN onboarding and data ferrying
  - Status monitoring and mission supervision
- This separation enables:
  - Clean modular design
  - Easy extension for new mission behaviors
  - Independent testing and debugging

Secure Wi-Fi Software Authentication (Onboarding Phase)

- When the UAV reaches a sensor node:
  - A Wi-Fi-based software authentication process is triggered
  - Only authenticated sensor nodes are allowed to onboard
- Prevents unauthorized data access
- Enables controlled and secure data ferrying in open environments

Context-Aware Mission Behavior

- UAV behavior adapts based on node type and terrain:
  - Tree-mounted nodes → hover at safe altitude
  - Ground nodes → controlled descent, hover, and climb
- Ensures safety while maintaining communication reliability

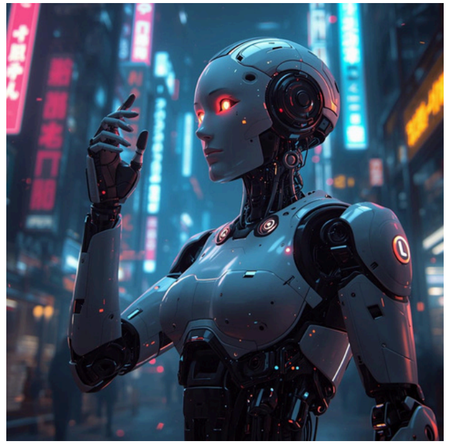
Key Requirements Addressed by the Solution

✓ Autonomous Operation

- Fully autonomous takeoff, navigation, hovering, and return
- No manual piloting required after mission start

✓ Intelligent Mission Planning

- Dynamic waypoint generation from mission payload
- Different behaviors based on ground type:
  - Tree → hover
  - Ground → descend, hover, climb
- Node-to-waypoint mapping for task tracking



✓ Reliability & Safety

- Uses ArduPilot's proven flight stack
- RTL for fail-safe recovery
- No forced landing at sensor nodes

✓ Modularity & Scalability

- ROS 2 node-based architecture
- Easy integration of:
  - New sensors
  - New behaviors
  - Additional UAVs

✓ Real-World Deployability

- Tested on Simulated setup first-hand
- Tested on real firmware
- Compatible with companion computers (Raspberry Pi, Jetson)

Design & Current Status

Feedback on Prototype

- Successful UAV build and calibration
- Stable flight after extensive:
  - PID tuning
  - Center of gravity adjustments
- Reliable waypoint execution in AUTO mode
- Correct handling of hover, descend, and climb behaviors
- ROS 2–MAVROS–ArduPilot integration validated
- Successful data onboarding to the onboard-computer

⚠ Challenges Encountered

- Multiple crashes during early PID tuning
- Sensitivity to payload weight and balance
- Timing and synchronization issues in mission execution
- Debugging real-time flight behavior vs planned logic

COMMERCIALIZATION / SUSTAINABILITY OF SOLUTION

AUTONOMOUS MULTI-SENSOR ENVIRONMENTAL MONITORING NODE

Customer Persona

- Environmental monitoring agencies (forestry, agriculture, disaster management).
- Organizations monitoring remote or hard-to-access areas.
- Research institutions requiring periodic data collection without fixed infrastructure.
- Smart agriculture operators with large-scale sensor deployments.

Market Size

- Growing demand for Wireless Sensor Networks (WSN) in environmental and scientific monitoring.
- Increasing adoption of UAV-assisted data collection for remote sensing.
- Applicable to local, regional, and global deployments, especially in developing regions.
- Market driven by need for low-cost, low-power monitoring solutions.

What is Unique?

- Adaptive power management engine that optimizes node and comparison between two compression algorithms.
- Application of the STELLAR algorithm to enhance CH election.
- Scheduled, time-sliced communication to minimize idle listening and reduce power consumption.
- Modular, embedded-ready implementation using real hardware, not simulation only.

Cost Recovering

- Low hardware cost using off-the-shelf ESP-based nodes.
- Reduced maintenance cost due to longer sensor battery life.
- One UAV can serve multiple sensor clusters, lowering operational costs.
- System can be offered as:
  - A deployment-based solution (one-time installation cost), or
  - A monitoring service (subscription-based data access).

Thank you!

DEMO :)!

UAV Testing Footage

Q & A